

TECHNICAL ASSIGNMENT

DockerForge — AI-Powered Dockerfile Generator

Full Stack + Agentic AI Developer

Overview

You will build an AI agent that takes a GitHub repository URL as input, analyzes the codebase, and generates a working Dockerfile (and optionally a docker-compose.yml) — one that actually builds and runs without errors.

This is not just a text generation task. The agent must verify its own output by building the Docker image and confirming the container starts correctly. If the build fails, the agent must read the error and retry.

Assignment Details

Field	Details
Role	Full Stack + Agentic AI Developer
Submission	GitHub repository link + short README
Tech Freedom	Any language or framework of your choice

Problem Statement

Build a tool called DockerForge with the following flow:

Step	What the Agent Does
1	Accept a public GitHub repository URL from the user
2	Clone the repository and scan its file structure
3	Generate a Dockerfile based on the analysis
4	Run: docker build . — capture and display the output



5	If build fails, read the error, reason about the fix, and retry (max 3 attempts)
6	Run: docker run — verify the container starts and responds
7	Display the final working Dockerfile to the user in the UI

Submission Instructions

1. Push all code to a public GitHub repository
2. Your project itself must be Dockerized — include a Dockerfile so your tool can be run with docker build and docker run out of the box
3. Your README must include: architecture diagram or explanation, setup instructions, which LLM provider you used and why, known limitations or edge cases
4. Record a 2-3 minute screen recording showing the tool working on the Medium-level test repo
5. Submit the GitHub link + video link to [your submission email/form]

Submissions without a working README or video will be scored lower.

Good luck. Build something that actually works.